1. SUPERMARKET SALES ANALYSIS

Project Overview

The **Supermarket Sales Data Analysis** project focuses on analyzing supermarket transaction data to understand sales trends, customer behavior, and product performance. The dataset includes information such as product categories, quantity purchased, sales amount, customer gender, branch location, payment methods, and customer ratings.

The main objective of this project is to use data analysis techniques to extract meaningful insights from the data. Python libraries such as **Pandas, Matplotlib, and Seaborn** are used to perform data cleaning, statistical analysis, and data visualization.

Through this analysis, the project helps identify top-selling products, preferred payment methods, peak sales days and hours, and branch performance. These insights can help businesses improve sales strategies, manage inventory efficiently, and enhance customer satisfaction.

Setup Instructions

Follow the steps below to set up and run the Supermarket Sales Data Analysis project.

Step 1: Install Python

Download and install Python (version 3.8 or higher) from the official website:
https://www.python.org/downloads/

Step 2: Install Anaconda (Recommended)

Install Anaconda, which includes Jupyter Notebook and essential data science tools.
Download from: https://www.anaconda.com/download

Step 3: Install Required Libraries

Open the terminal or Anaconda Prompt and install the required Python libraries using the following command:

```
pip install pandas matplotlib seaborn numpy jupyter
```

Step 4: Prepare the Dataset

Download the dataset file supermarket_sales.csv and place it in the project folder where the Jupyter Notebook file is located.

Step 5: Launch Jupyter Notebook

Open the terminal or Anaconda Prompt and run:

```
jupyter notebook
```

Step 6: Run the Project

Open the notebook file (for example, *Supermarket_Sales_Analysis.ipynb*) and run the cells sequentially to load the dataset, perform analysis, and generate visualizations.

Code Structure

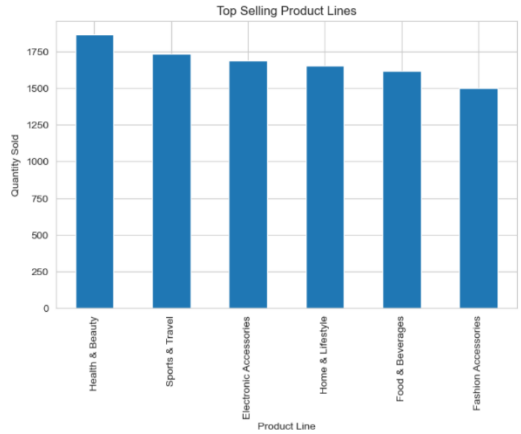
The project code is organized into multiple sections to make the analysis clear, structured, and easy to understand. Each section performs a specific task in the data analysis process.

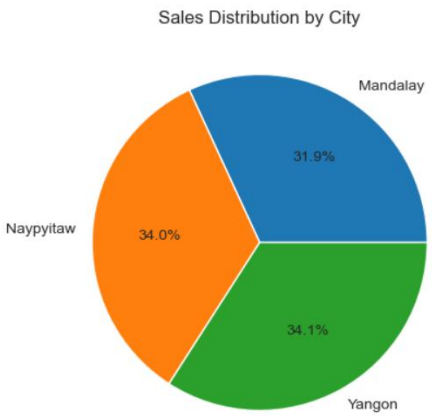
1. Import Libraries

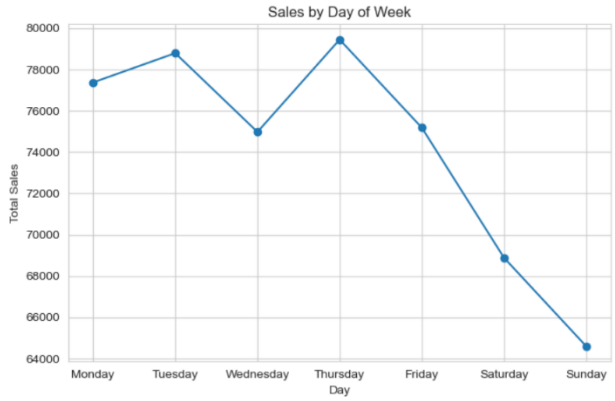
This section imports the required Python libraries such as Pandas for data manipulation, Matplotlib and Seaborn for data visualization.

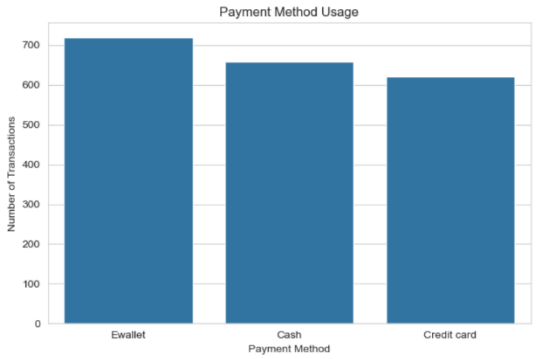
2. Load Dataset
The dataset (supermarket_sales.csv) is loaded into a Pandas DataFrame to begin the analysis.
3. Dataset Exploration
Basic information about the dataset is examined, including the number of rows and columns, column names, and data types.
4. Missing Values Check
The dataset is checked for any missing or null values to ensure data quality before performing analysis.
5. Data Cleaning and Feature Engineering
The Date and Time columns are converted into proper datetime format. Additional features such as Month, Day, and Hour are extracted to help analyze sales trends over time.
6. Basic Sales Statistics
Key statistics such as total sales, average transaction value, and total items sold are calculated to understand overall performance.
7. Sales Analysis by Product Line
The code analyzes which product categories have the highest sales and visualizes them using bar charts.
8. Sales Analysis by City
Sales are grouped by city to understand how revenue is distributed across different locations.
9. Sales by Day of the Week
This section identifies which days generate the most sales.
10. Payment Method Analysis
The distribution of different payment methods (cash, credit card, e-wallet) is analyzed.
11. Customer Rating Analysis
Customer ratings are visualized to understand overall satisfaction levels.
12. Correlation Analysis
A heatmap is created to identify relationships between numerical variables such as price, quantity, tax, and total sales.
13. Gender-based Analysis
Customer spending and product preferences are analyzed based on gender.
14. Time-based Sales Analysis
Sales trends are analyzed based on hourly transactions to identify peak shopping hours.
15. Branch Performance Analysis
Sales performance and customer ratings are compared across different supermarket branches.
16. Advanced Visualization
A heatmap is generated to analyze sales patterns based on both day of the week and hour of the day.
This structured approach ensures the project flows logically from data loading → cleaning → analysis → visualization → insights.

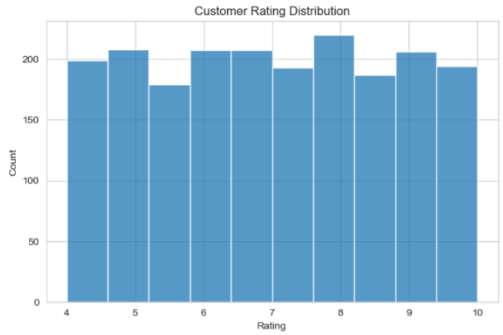
Visual Documentation

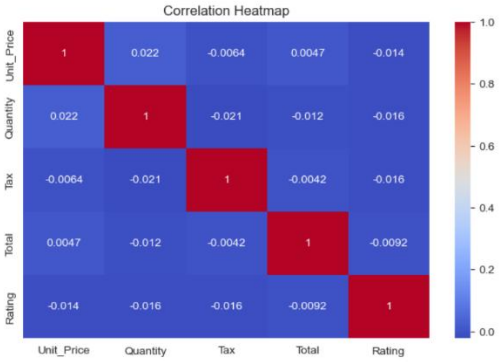

Top Selling Product Lines
Quantity Sold
1750
1500
1250
1000
750
500
250
0
Health & Beauty
Sports & Travel
Electronic Accessories
Home & Lifestyle
Food & Beverages
Fashion Accessories
Product Line

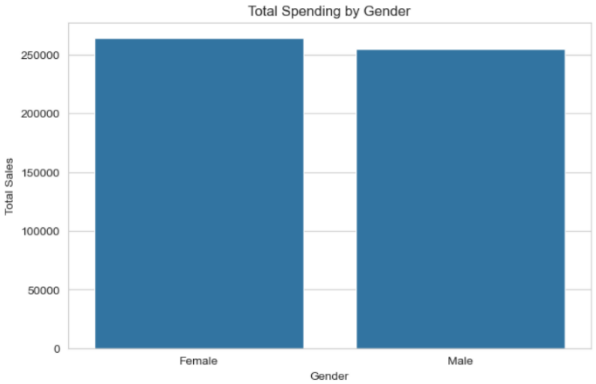

Sales Distribution by City
Mandalay
31.9%
Naypyitaw
34.0%
34.1%
Yangon

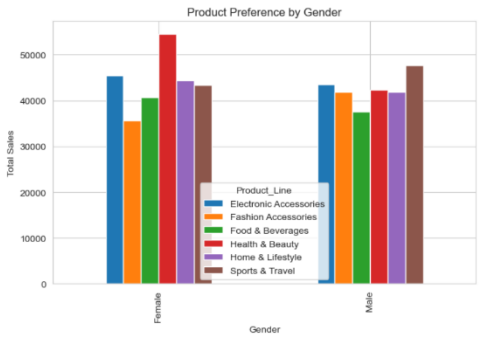

Sales by Day of Week
Total Sales
80000
78000
76000
74000
72000
70000
68000
66000
64000
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day


Payment Method Usage
Number of Transactions
700
600
500
400
300
200
100
0
Ewallet
Cash
Credit card
Payment Method

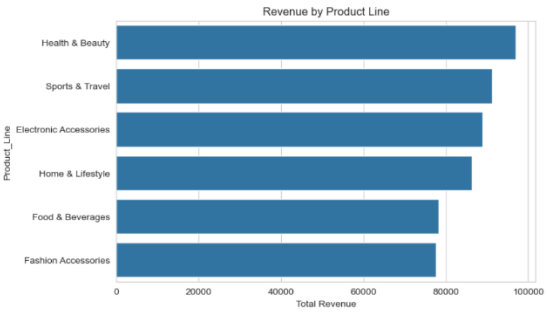

Customer Rating Distribution
Count
200
150
100
50
0
4
5
6
7
8
9
10
Rating

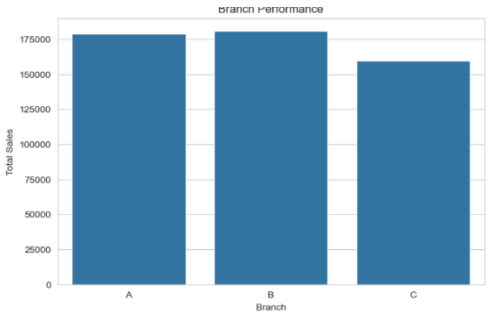

Correlation Heatmap
Unit_Price
Quantity
Tax
Total
Rating
1
0.022
-0.0064
0.0047
-0.014
0.022
1
-0.021
-0.012
-0.016
-0.0064
-0.021
1
-0.0042
-0.016
0.0047
-0.012
-0.0042
1
-0.0092
-0.014
-0.016
-0.016
-0.0092
1
1.0
0.8
0.6
0.4
0.2
0.0

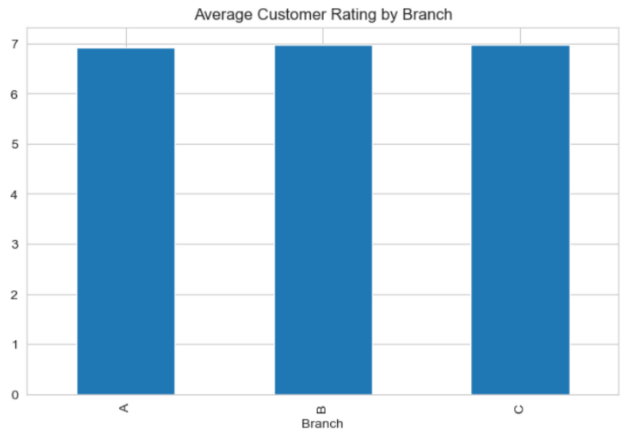

Total Spending by Gender
Total Sales
250000
200000
150000
100000
50000
0
Female
Male
Gender

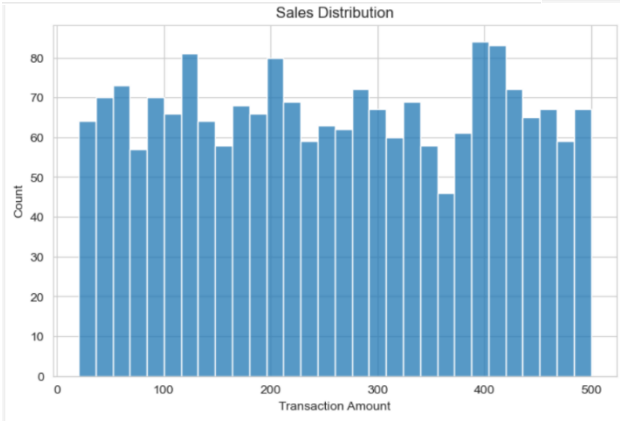

Product Preference by Gender
Total Sales
50000
40000
30000
20000
10000
0
Product_Line
Electronic Accessories
Fashion Accessories
Food & Beverages
Health & Beauty
Home & Lifestyle
Sports & Travel
Female
Male
Gender

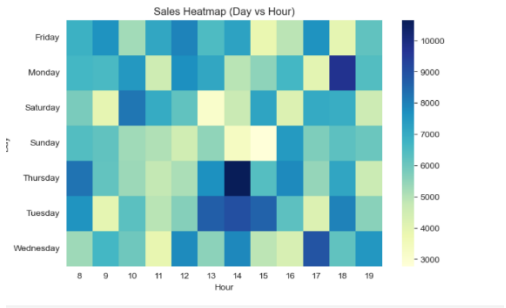
Sales by Hour
Total Sales
48000
46000
44000
42000
40000
38000
8
10
12
14
16
18
Hour


Revenue by Product Line
Product_Line
Health & Beauty
Sports & Travel
Electronic Accessories
Home & Lifestyle
Food & Beverages
Fashion Accessories
0
20000
40000
60000
80000
100000
Total Revenue









Technical Details

The Supermarket Sales Data Analysis project is implemented using the Python programming language and several data analysis and visualization libraries. These tools help process the dataset, perform statistical analysis, and generate graphical insights.

Programming Language

- Python 3.x is used as the primary programming language for data analysis and visualization.

Libraries Used

- Pandas – Used for data loading, cleaning, manipulation, and analysis.
- Matplotlib – Used to create basic charts and visualizations such as bar charts, line plots, and pie charts.
- Seaborn – Used for advanced statistical visualizations such as heatmaps, histograms, and styled charts.

Development Environment

- Jupyter Notebook is used to write and execute the code step-by-step, making it easier to analyze data interactively and visualize results.

Dataset Format

- The dataset is stored in CSV (Comma-Separated Values) format and is loaded using Pandas.

Data Processing Techniques

- Data type conversion (Date and Time to datetime format)
- Feature extraction (Month, Day, Hour)
- Data aggregation using groupby()
- Statistical calculations (sum, mean, correlation)

Visualization Techniques

- Bar charts for categorical comparisons
- Line charts for time-based trends
- Pie charts for distribution analysis

- Histograms for data distribution
- Heatmaps for correlation and time-based sales analysis

These technical components work together to transform raw supermarket transaction data into meaningful insights and visual representations.

Testing Evidence

Testing was performed to ensure that the **Supermarket Sales Data Analysis** project works correctly and produces accurate results. Each section of the code was executed step-by-step in **Jupyter Notebook** to verify that the dataset loads properly, the analysis runs without errors, and the visualizations are generated correctly.

1. Dataset Loading Test

The dataset supermarket_sales.csv was successfully loaded using the Pandas read_csv() function. The df.head() command was used to confirm that the data was imported correctly.

2. Data Exploration Test

Commands such as df.shape, df.columns, and df.info() were executed to check the structure of the dataset. This confirmed the number of rows, columns, and data types.

3. Missing Value Test

The df.isnull().sum() function was used to verify that the dataset does not contain missing values that could affect the analysis.

4. Data Conversion Test

The **Date** and **Time** columns were converted to datetime format using pd.to_datetime() to ensure proper time-based analysis. New features such as **Month, Day, and Hour** were extracted successfully.

5. Statistical Calculation Test

Key calculations such as **total sales**, **average transaction value**, and **total items sold** were computed and verified.

6. Visualization Test

All visualizations including **bar charts, pie charts, histograms, line charts, and heatmaps** were generated successfully without errors. These charts clearly displayed the sales patterns and relationships within the dataset.

7. Output Verification

The results produced by the analysis were checked to ensure that the insights matched the dataset values and calculations.

Overall, testing confirmed that the project runs correctly and produces accurate analytical results and visualizations.